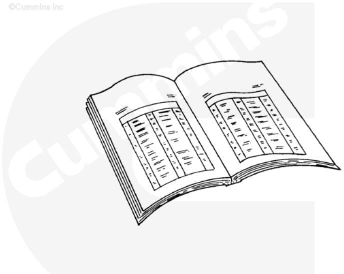
Trainings

Leak Test

This test should **only** be performed when the oil flow of the Centinel™ valve needs to be verified. This test does **not** test for external leaks.

This test is divided into four portions. Refer to the portion of the test that corresponds to the Centinel™ installation that is being serviced.

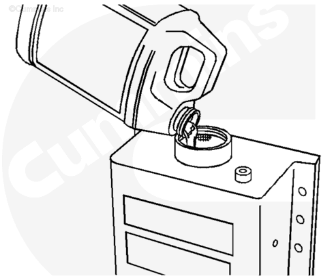


ck800wa

Dual-Piston

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



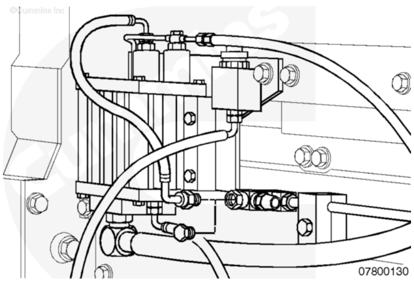
05100048

Fill the make-up tank to the full mark with clean engine oil.

With the engine **not** running, disconnect the oil burn hose at the fuel return line fitting.

Plug the fuel return line to prevent fuel leakage.

If a make-up system is being serviced, disconnect the clean oil hose at the block fitting. Do **not** disconnect the used oil supply hose that is attached to a fitting in the oil rifle.

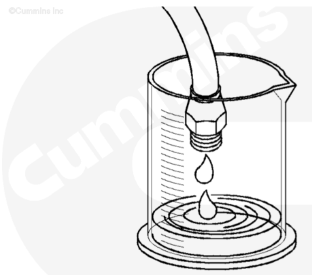


07800130

Place a graduated cylinder or other measuring container under the disconnected hoses to capture the oil.

Start the engine.

Some oil may initially flow out of the oil drain (or clean oil) hose. This oil should be discarded before continuing.

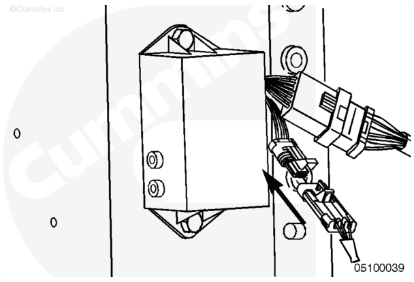


07800129

Monitor both hoses for two minutes. If oil continues to flow out of either hose, there is an internal leak and the valve **must** be rebuilt or replaced.

After the first discharge of oil is discarded, put the Centinel™ system in service mode. Refer to Procedure 007-999 (/qs3/pubsys2/xml/en/procedures/96/96-007-999.html).

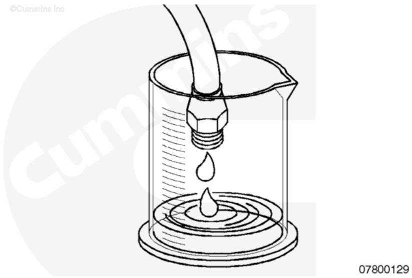
When the service mode plug is installed, the Centinel™ valve will cycle automatically.



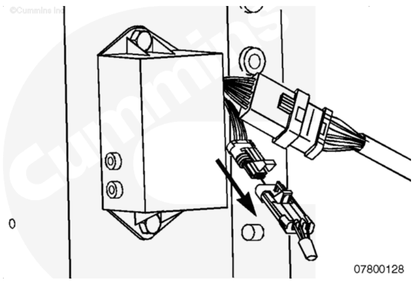
Use a graduated cylinder or other measuring container to capture the oil.

One full cycle will consist of one discharge of oil measuring 34 ml +/- 4 ml from the oil burn hose and four discharges totaling 150 ml +/- 10 ml from the oil drain (or clean oil) hose. Capturing more than one cycle and taking an average of the oil output is recommended to minimize measurement error. These flow values are for both burn only and make-up systems.

The control valve **must** be rebuilt or replaced if the oil discharge is **not** within specification.



Shut down the engine and restore the hoses to the original configuration.

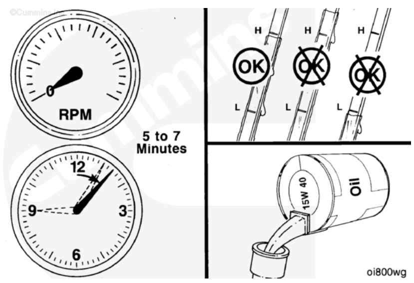


⚠ CAUTION ⚠

Overfilling the engine with oil can cause damage to the engine.

Wait five to seven minutes for the oil to drain back to the pan.

Check the oil level on the dipstick and the oil level in the make-up tank, if equipped, and add clean oil as necessary.



If the Centinel™ valve is found to be operating properly and too much oil is being used from the crankcase, verify regular engine oil consumption of the engine.

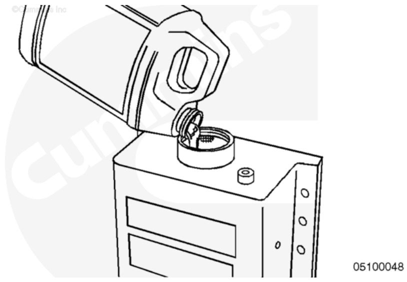
The Centinel™ system **must** first be isolated from the engine. This is done by disconnecting all Centinel™ oil lines from the engine and plugging all open ports.

Refer to the Lubricating Oil Consumption Excessive symptom chart in the appropriate troubleshooting and repair manual for further troubleshooting information.

Single-Piston

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



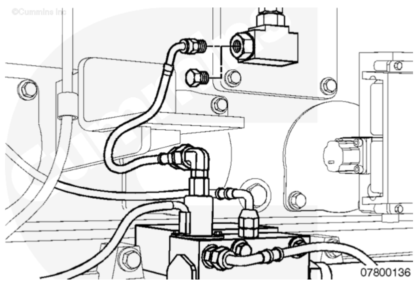
Note : Some single piston Centinel™ systems can use a service plug and do not need to be cycled manually.

Fill the make-up tank to the full mark with clean engine oil.

With the engine **not** running, disconnect the oil burn hose at the fuel return line fitting or integrated fuel system module (IFSM) port.

Plug the open port to prevent fuel leakage.

If a make-up system is being serviced, disconnect the clean oil hose at the block fitting. Do **not** disconnect the used oil supply hose that is attached to a fitting in the oil rifle.



Place a graduated cylinder or other measuring container under the disconnected hoses to capture the oil.

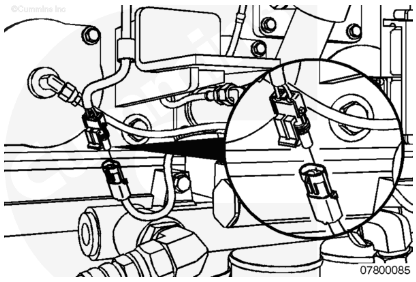
Start the engine.

Some oil will initially flow out of the oil drain (or clean oil) hose. This oil should be discarded before continuing.

Monitor both hoses for two minutes. If oil continues to flow out of either hose, there is an internal leak and the valve **must** be rebuilt or replaced.

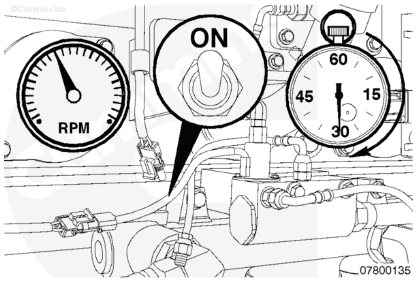
Disconnect the solenoid from the engine harness and connect a switchable power source of the proper voltage to the solenoid leads.

The power **must** be in the OFF position when it is initially connected to the solenoid.



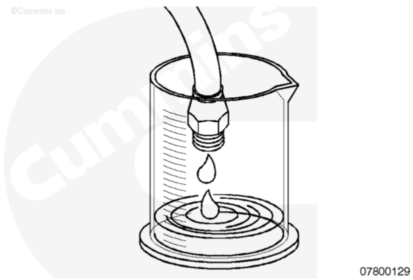
Cycle the power to the solenoid at 30 second intervals: power ON for 30 seconds, then OFF for 30 seconds.

This will give a complete cycle of 60 seconds.



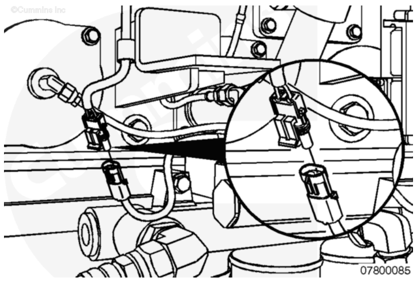
Cycle the solenoid several times. Each full cycle should discharge 17 ml +/- 3 ml from each oil line. Divide the total oil volume by the number of cycles to get an average reading.

The control valve **must** be rebuilt or replaced if the oil discharge is **not** within specification.

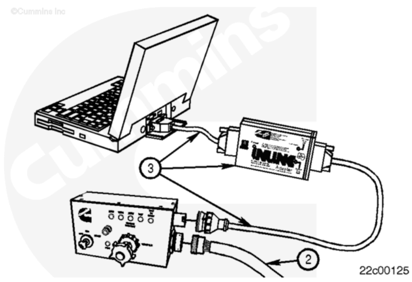


Shut down the engine and restore the hoses to the original configuration.

Reconnect the solenoid to the engine harness.

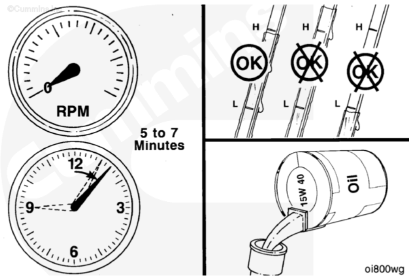


Use the Cummins INSITE™ electronic service tool to clear all inactive Centinel™ fault codes in the ECM. Refer to the appropriate INSITE™ User's Manual.



⚠ CAUTION ⚠

Overfilling the engine with oil can cause damage to the engine.



Wait five to seven minutes for the oil to drain back to the pan.

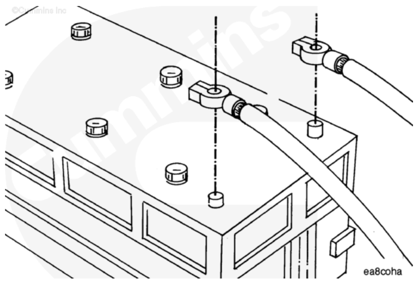
Check the oil level on the dipstick and the oil level in the make-up tank, if equipped, and add clean engine oil as necessary.

If the Centinel™ valve is found to be operating properly and too much oil is being used from the crankcase, verify regular engine oil consumption of the engine.

The Centinel™ system **must** first be isolated from the engine. This is done by disconnecting all Centinel™ oil lines from the engine and plugging all open ports.

Refer to the Lubricating Oil Consumption Excessive symptom chart in the appropriate troubleshooting and repair manual for further troubleshooting information.

Preparatory Steps



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

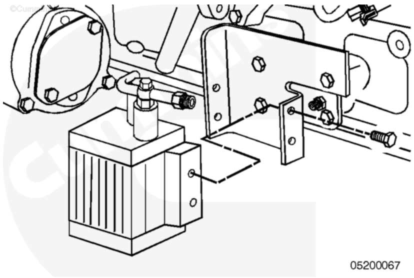
Disconnect the battery cables

Remove the Centinel™ oil lines. Refer to Procedure 007-088 (/qs3/pubsys2/xml/en/procedures/96/96-007-088-tr.html).

Remove

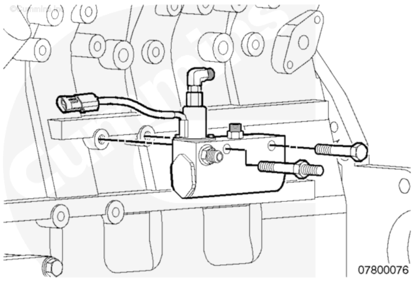
L10 and M11 Engines

Remove the four M6 X 1 - 20-mm bolts and the oil control valve assembly.



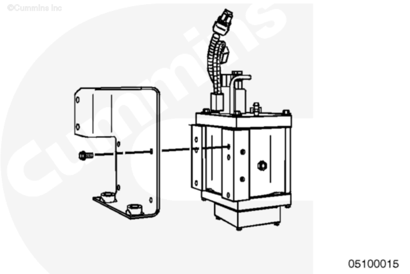
ISM Engines

Remove the Centinel™ oil control valve from the ISM engine block.



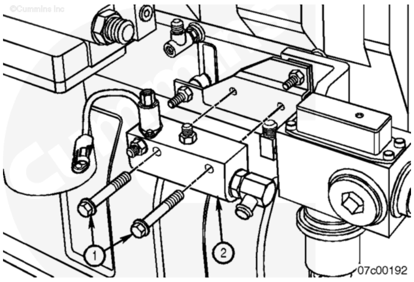
N14 Engines

Remove the Centinel™ oil control valve and bracket assembly.



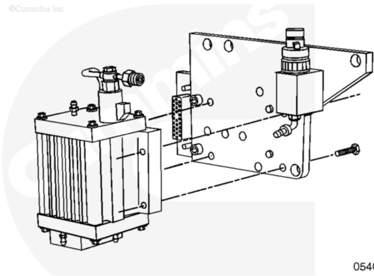
ISX Engines

Remove the Centinel™ oil control valve (2) from its bracket.



K19 Engines

Remove the Centinel™ oil control valve from the oil control valve mounting bracket.



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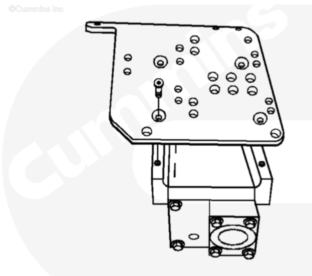
K38 and K50 Engines

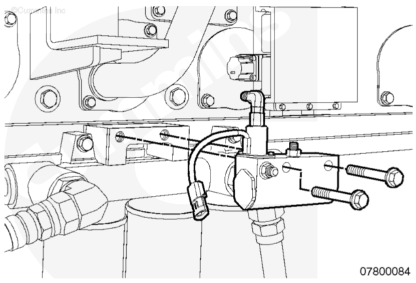
Remove the Centinel™ oil control valve from the mounting bracket.



05600050

QSK45 and QSK60

Remove the Centinel™ oil control valve from the mounting bracket.



07800084

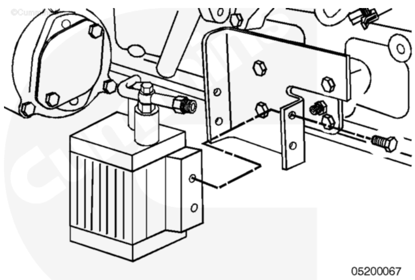
Install

L10 and M11 Engines

Using the four M6 X 1 - 20-mm bolts provided, install the oil control valve assembly to the oil control valve mounting bracket.

Tighten the four mounting bolts.

Torque Value: 14 n•m [124 in-lb]

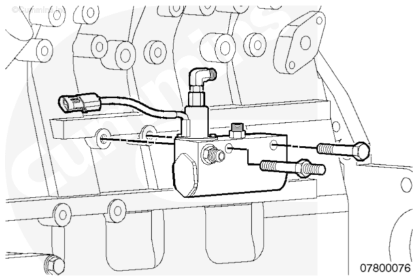


05200067

ISM Engines

Install the Centinel™ oil control valve on the ISM engine block using the one studded M10 x 1.5 and one flange head M10 x 1.5 capscrews. The capscrews insert through the two valve body mounting holes with the M10 studded capscrew toward the front of the engine. Tighten the two capscrews.

Torque Value: 47 n•m [35 ft-lb]

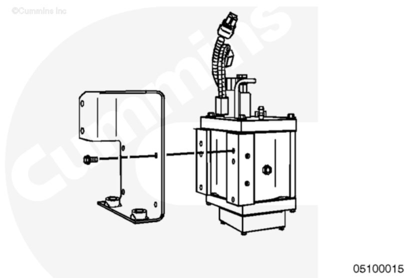


N14 Engines

Locate and remove the mounting bracket from the package.

Install the oil control valve to the bracket using the four M6 X 1 - 20-mm bolts provided.

Torque Value: 14 n•m [124 in-lb]

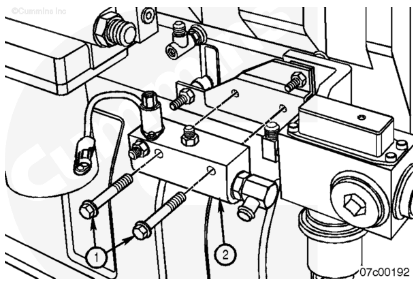


ISX Engines

Install the Centinel™ oil control valve (2) to its bracket.

Tighten the two capscrews (1).

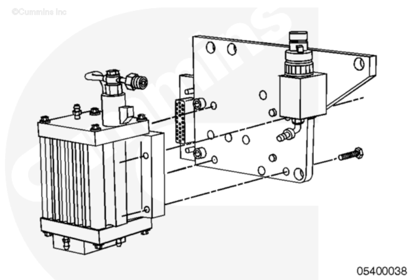
Torque Value: 47 n•m [35 ft-lb]



K19 Engines

Use the four M6-1 x 20-mm flat head bolts provided to install the oil control valve onto the oil control valve mounting bracket.

Torque Value: 14 n•m [124 in-lb]

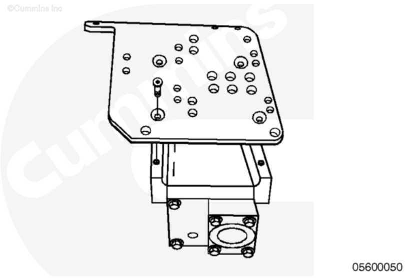


K38 and K50 Engines

Locate and remove the oil control valve and the oil control valve mounting bracket from the upfit kit.

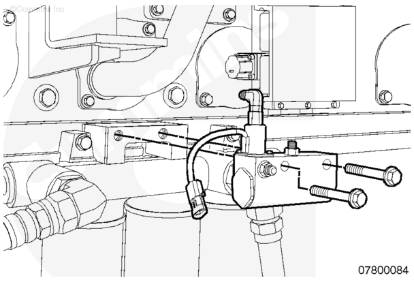
Install the oil control valve onto the oil control valve mounting bracket using the four M6-1 x 20-mm capscrews provided.

Torque Value: 14 n•m [124 in-lb]



QSK45 and QSK60

Install the Centinel™ oil control valve on the mounting bracket using the two M10-1.5 x 60 hex flange head capscrews. The capscrews insert through the two valve body mounting holes. Position the valve with the solenoid toward the front of the engine and the make-up line inlet toward the rear of the engine. Tighten the two capscrews.



Torque Value: 47 n•m [35 ft-lb]

Finishing Steps

Install the Centinel™ oil lines. Refer to Procedure 007-088 (/qs3/pubsys2/xml/en/procedures/96/96-007-088.html).

Last Modified: 20-Apr-2004